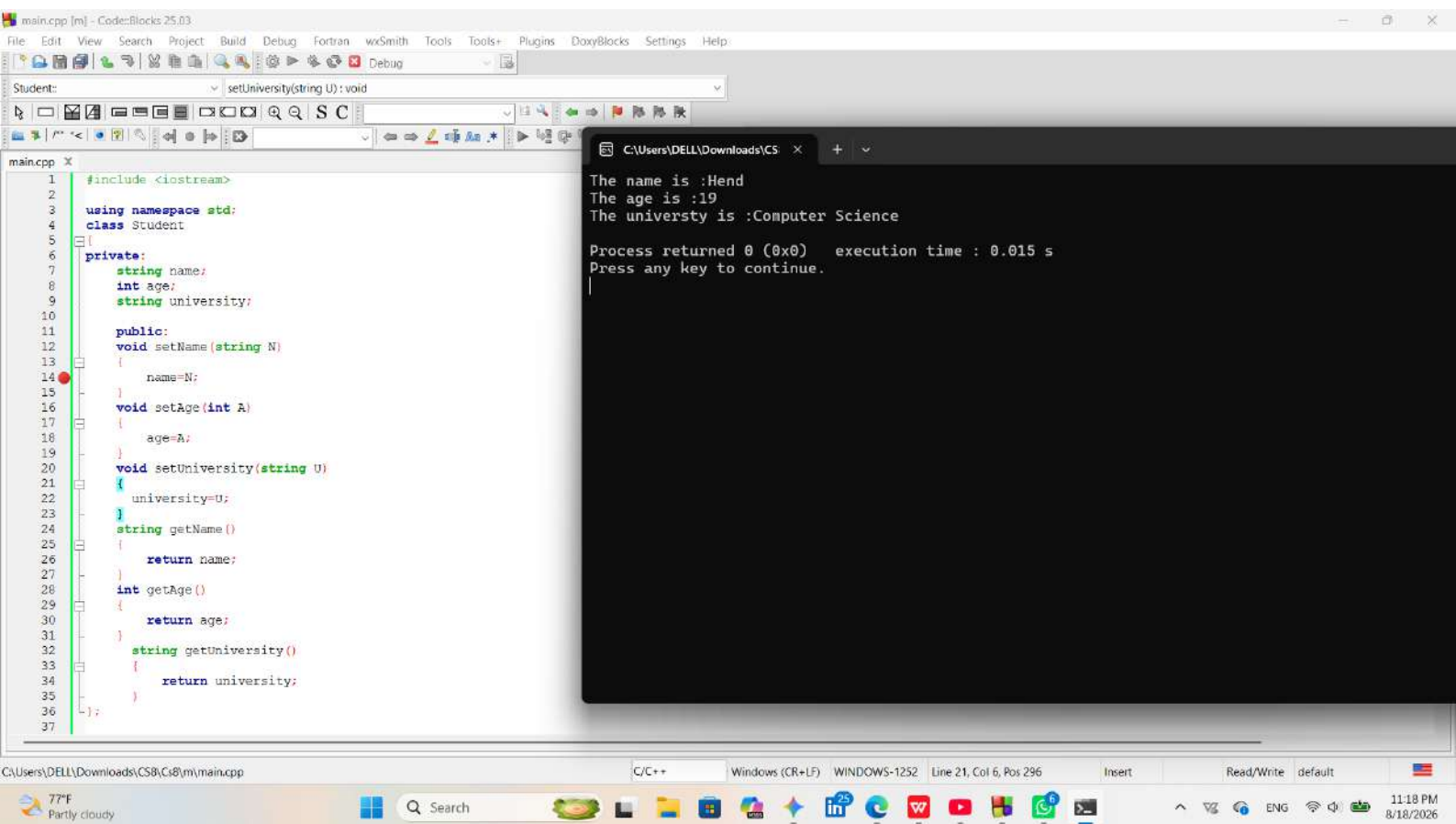




main.cpp [m] - Code::Blocks 25.03

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Student: setUniversity(string U): void

main.cpp

```
16 void setAge(int A)
17 {
18     age=A;
19 }
20 void setUniversity(string U)
21 {
22     university=U;
23 }
24 string getName()
25 {
26     return name;
27 }
28 int getAge()
29 {
30     return age;
31 }
32 string getUniversity()
33 {
34     return university;
35 }
36 };
37
38
39 int main()
40 {
41     Student data;
42     data.setName("Hend");
43     data.setAge(19);
44     data.setUniversity("Computer Science");
45     cout<<"The name is :"<<data.getName()<<endl;
46     cout<<"The age is :"<<data.getAge()<<endl;
47     cout<<"The university is :"<<data.getUniversity()<<endl;
48
49
50 }
51
```

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```
The name is :Hend
The age is :19
The university is :Computer Science

Process returned 0 (0x0)   execution time : 0.015 s
Press any key to continue.
```

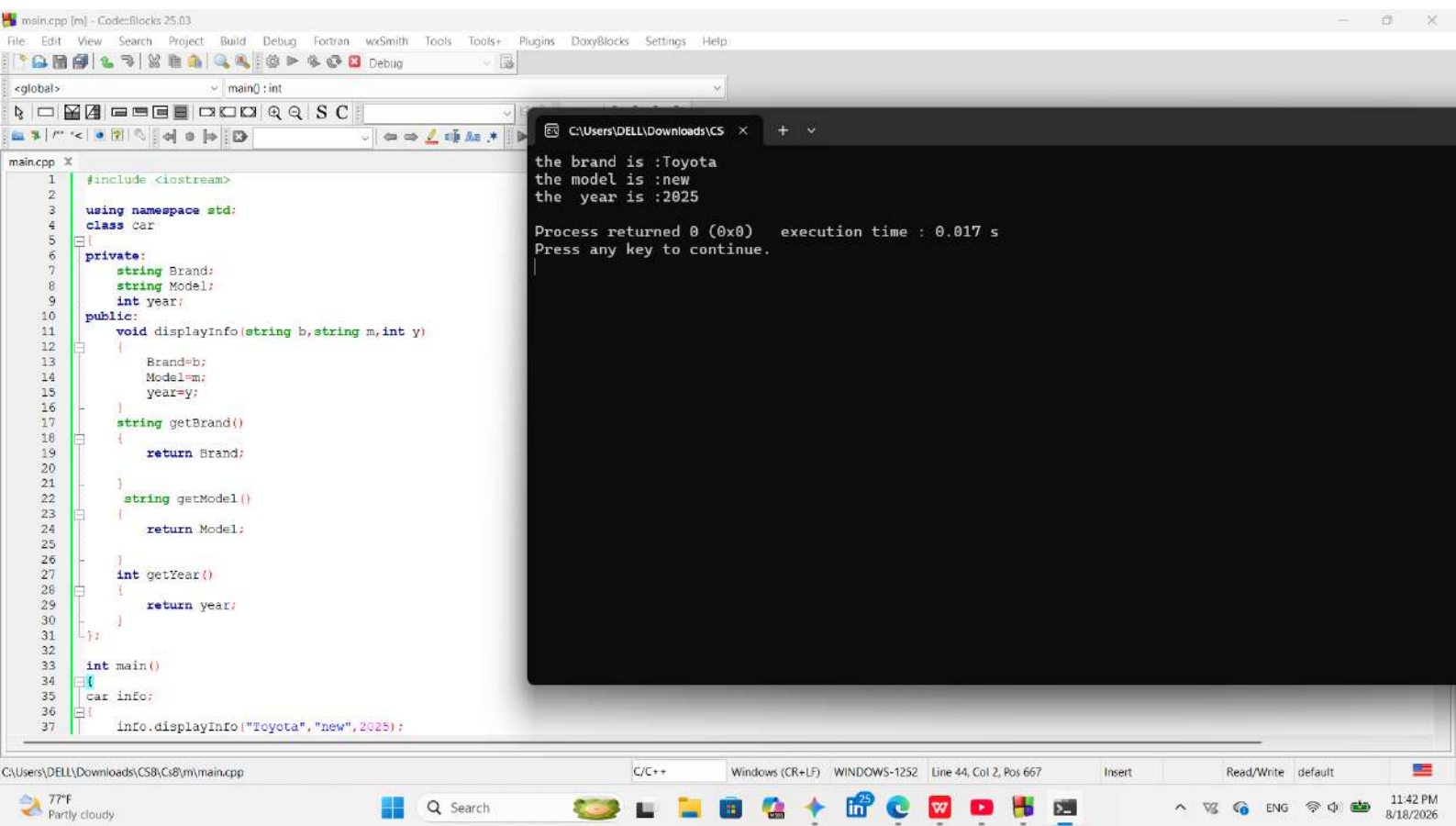
C:\Users\DELL\Downloads\CS8\l\m\main.cpp

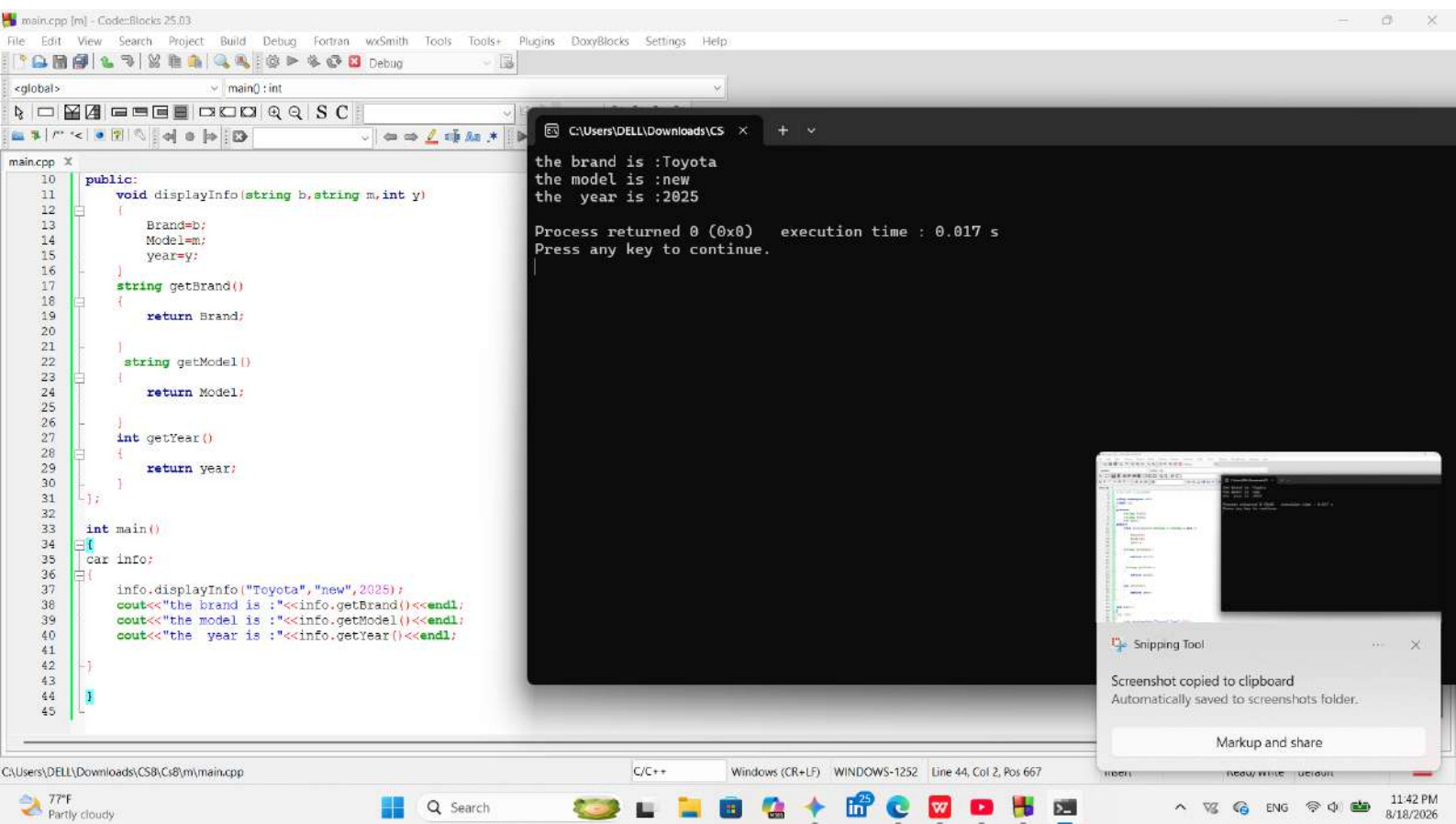
C/C++ Windows (CR+LF) WINDOWS-1252 Line 21, Col 5, Pos 296 Insert Read/Write default

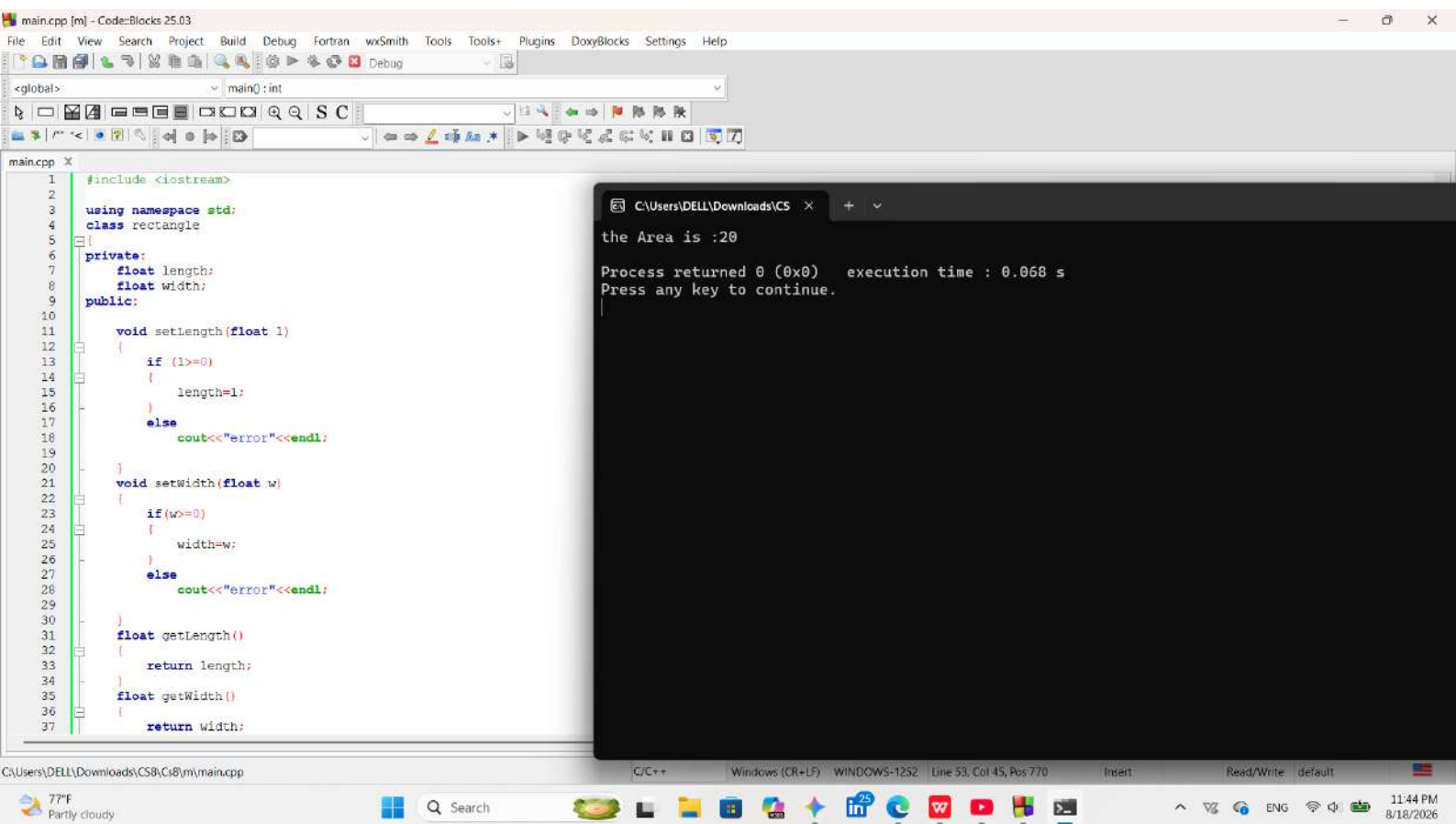
77°F Partly cloudy

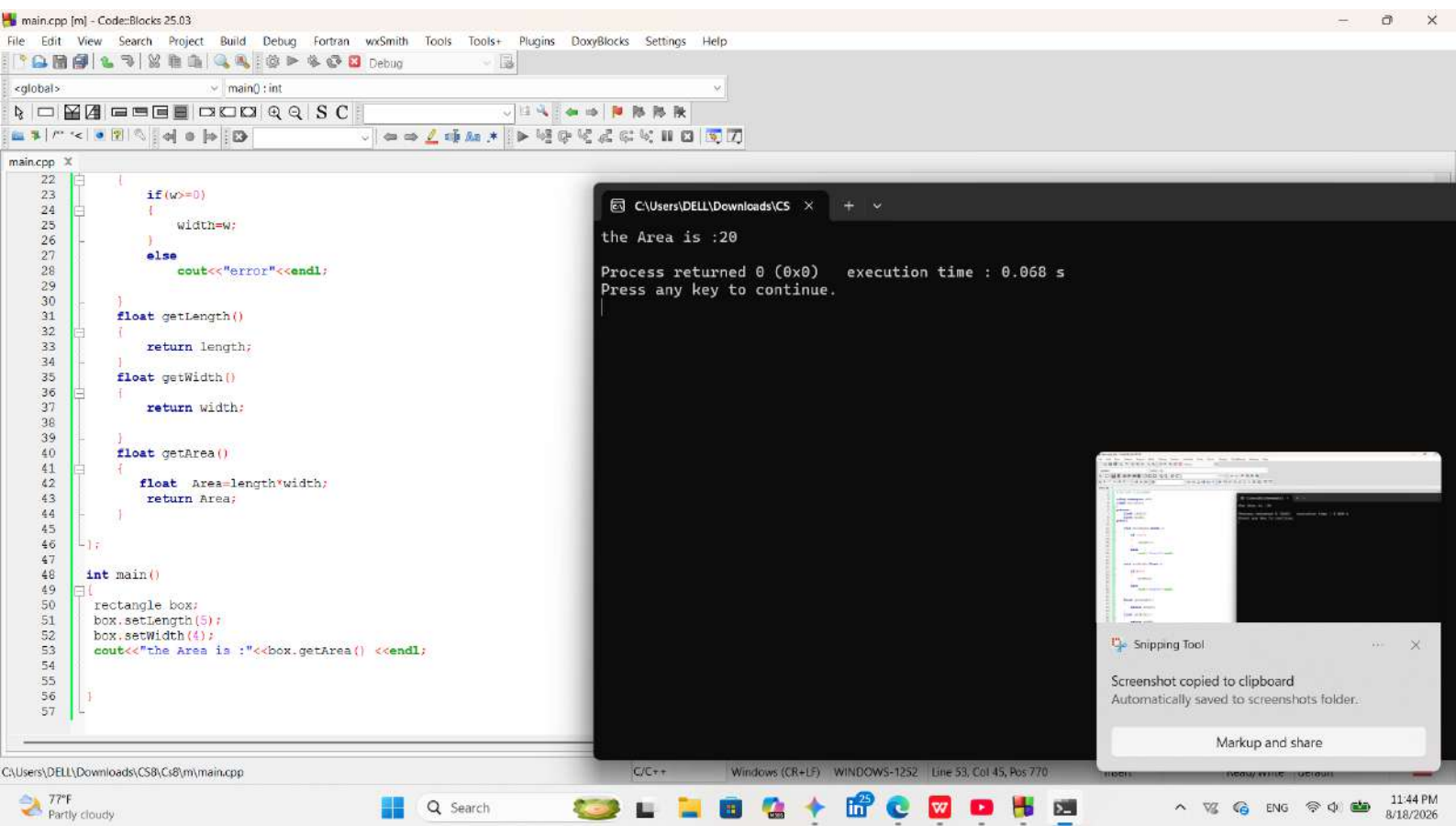
Search

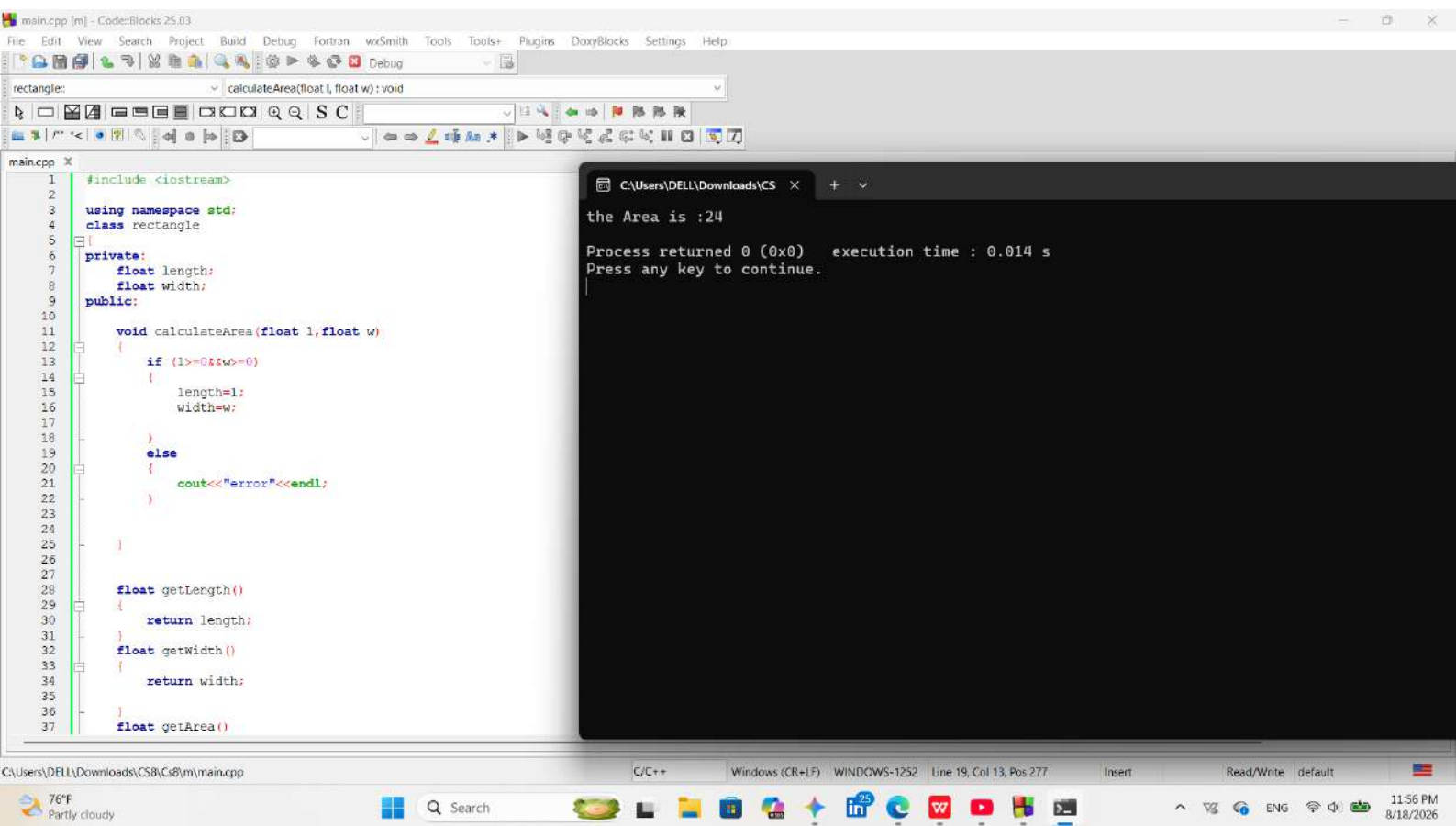
11:18 PM 8/18/2026

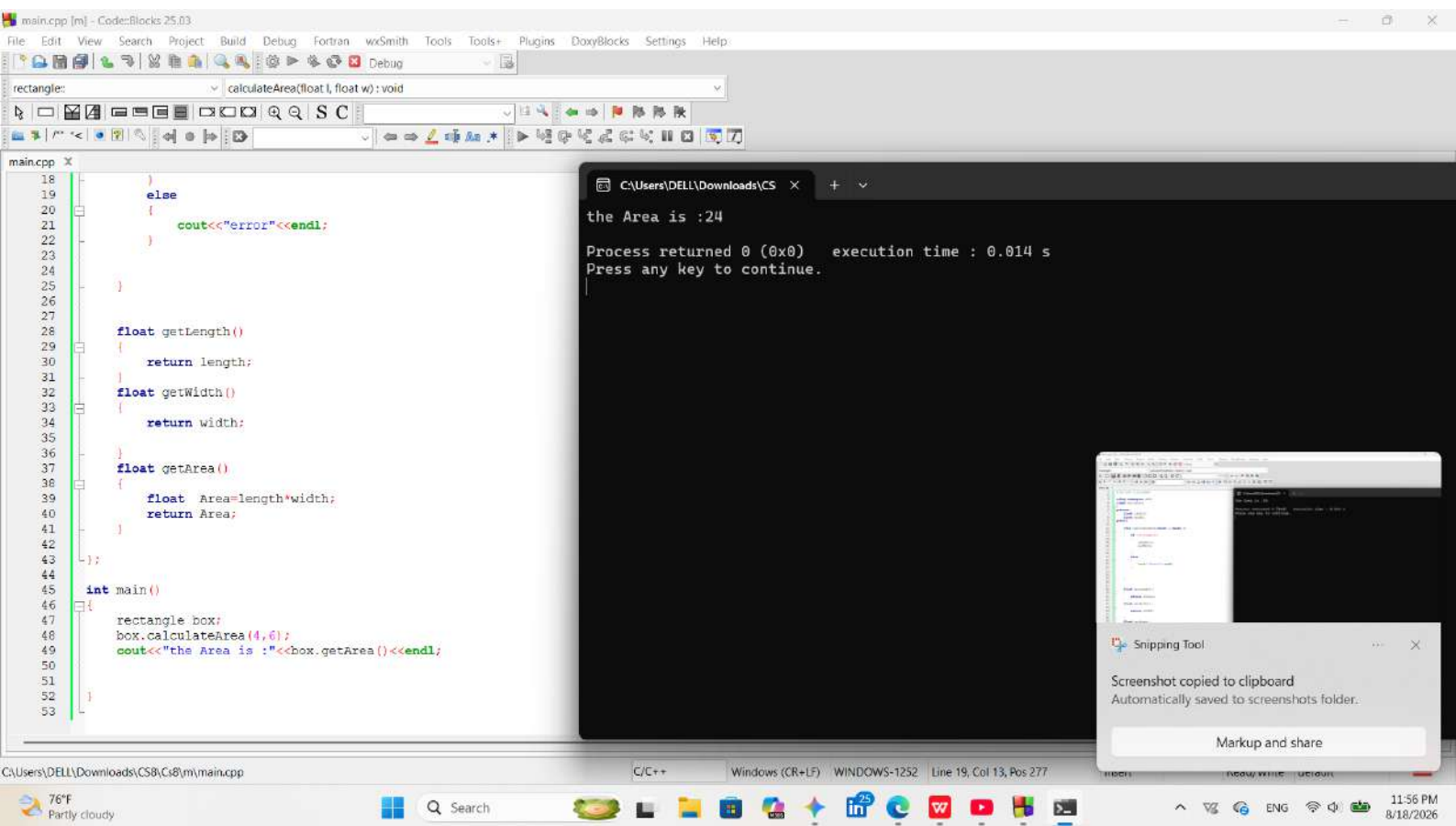


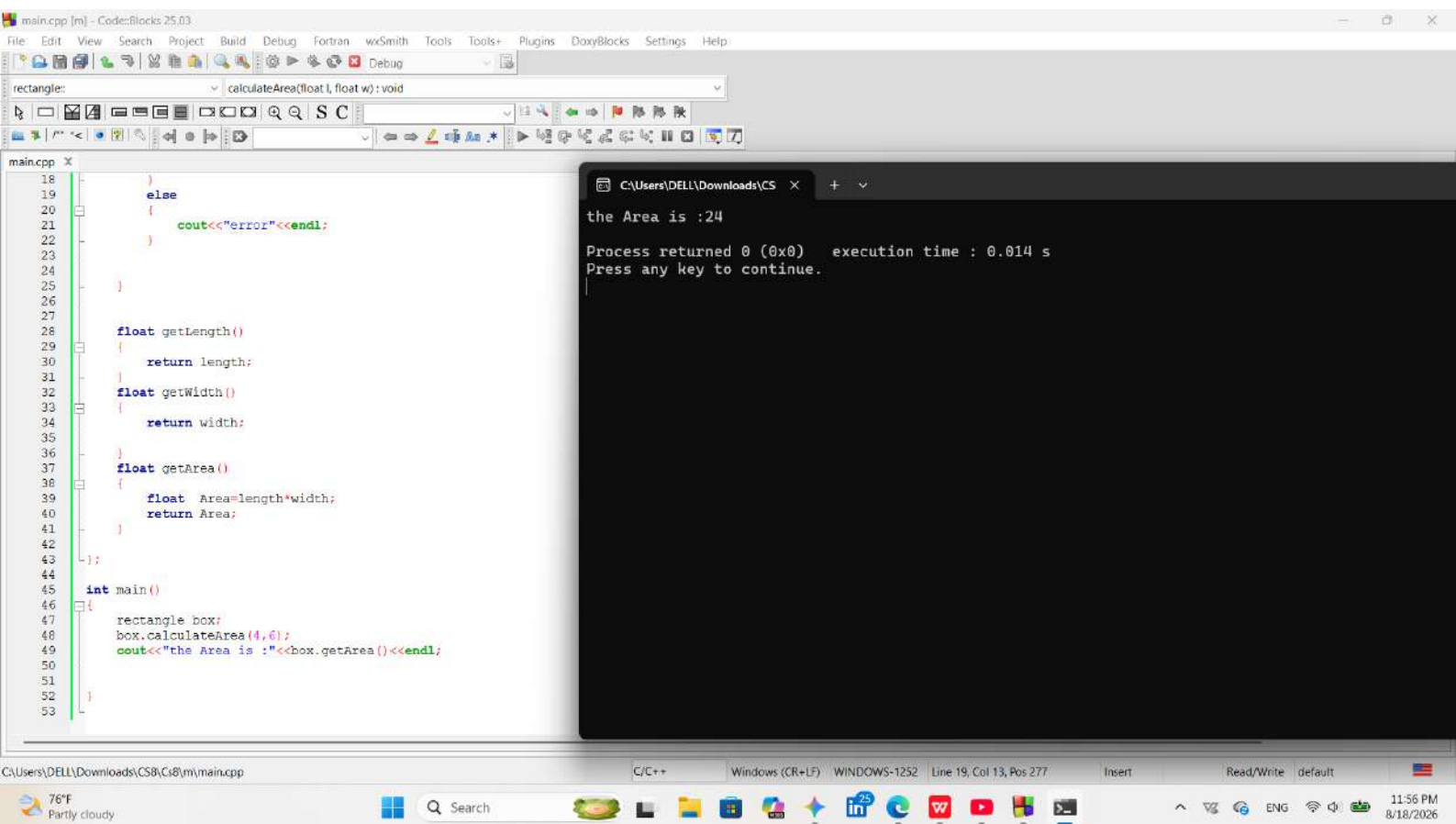












main.cpp [m] - Code::Blocks 25.03

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BankAccount: setAccountNumber(int N): void

main.cpp

```
1 #include <iostream>
2
3 using namespace std;
4 class BankAccount
5 {
6 private:
7     int accountNumber;
8     string balance;
9 public:
10    void setAccountNumber(int N)
11    {
12        if (N>0)
13        {
14            accountNumber=N;
15        }
16        else
17            cout<<"Error,please enter positive number"<<endl;
18    }
19    void setBalance(string B)
20    {
21        balance=B;
22    }
23    int getAccountNumber()
24    {
25        return accountNumber;
26    }
27    string getBalance()
28    {
29        return balance;
30    }
31 }
32
33
34 };
35
36 int main()
37 {
```

C:\Users\DELL\Downloads\CS

The accountNumber is:32456
The balance is:Satisfied

Process returned 0 (0x0) execution time : 0.018 s
Press any key to continue.

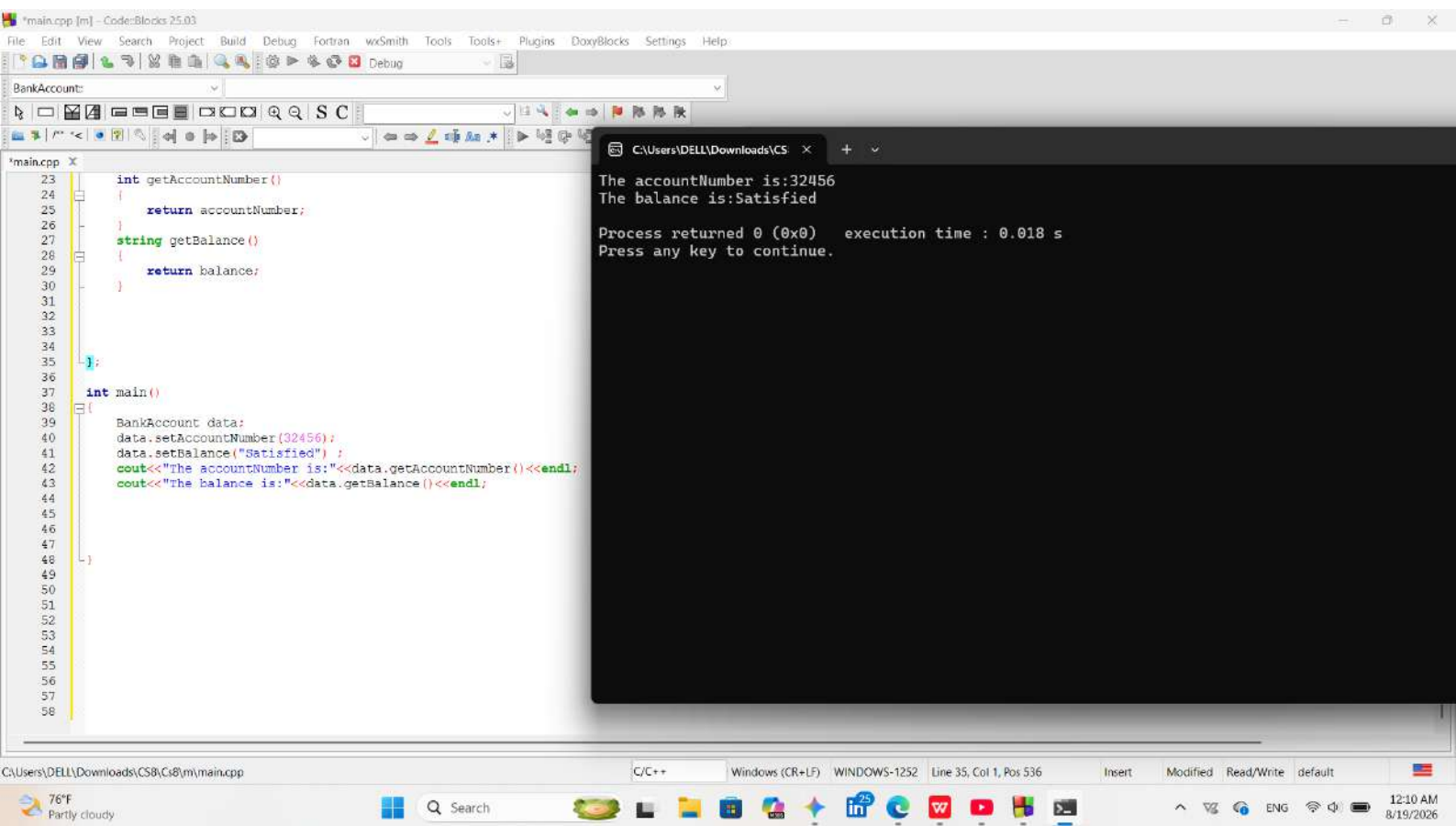
C:\Users\DELL\Downloads\CS\8\m\main.cpp

C/C++ Windows (CR+LF) WINDOWS-1252 Line 14, Col 23, Pos 231 Insert Read/Write default

76°F Partly cloudy

Search

12:10 AM 8/19/2026



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Task 10:

Class is general (الشيء العام)

يكون جوازا

يكون الموجهة لشيء من كذا

class

Class and Object in OOP

1. Class Definition

A Class is a user-defined blueprint, template, or prototype from which objects are created. It defines a set of attributes (variables) and methods (functions) that the created objects will have.

Example: A Car class that defines properties like color and brand, and behaviors like drive().

2. Object Definition and the Relationship Between Them

An Object is an instance of a class. It is a real-world entity that has a state (attributes) and behavior (methods) allocated in memory.

The Relationship: The class acts as the general blueprint or design, while the object is the concrete, physical realization of that blueprint. Multiple objects can be created from a single class, sharing its structure but holding their own unique data values.

Task 11:

Advantages of Inheritance in OOP

1. Code Reusability (إعادة استخدام الكود)

Description: Inheritance allows a new class (derived/child class) to adopt the properties and methods of an existing class (base/parent class). This eliminates the need to rewrite the same code from scratch, saving time and effort.

Example: If you have a base class Animal with a method eat(), any child class like Dog or Cat automatically inherits the eat() method without rewriting it.

2. Maintainability and Extensibility (سهولة الصيانة والتطوير)

Description: Because common code is centralized in the parent class, making changes or fixing bugs in one place automatically updates all child classes. Furthermore, you can easily extend functionality in the child classes by adding new features or overriding existing ones without breaking the parent code.

Page 2 of 2281 wordsEnglish (United States)Accessibility: Good to go

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Focus

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